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Investigating the Impact of Climate Change on the Caspian Sea Level and Ecological Conditions

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Abstract: The Caspian Sea, the largest inland body of water in the world, is distinguished by its geographical location, natural resources, and unique characteristics. Over many years, the ecosystem of the Caspian has been negatively impacted by industrial and household wastewater, the oil industry, maritime transport, and waste from tourism facilities along its coast. Furthermore, the ecological problems caused by the long-term fluctuations in the sea's water level, coupled with the current threat posed by climate change, present significant and diverse challenges to the region's water resources and both the coastal and marine ecosystems of the Caspian Sea basin. The ecological condition, the fluctuations in water levels, and the overall status of the Caspian are among the critical contemporary issues facing the region. In the context of climate variability, the Caspian Sea is confronted with complex and pressing challenges that require urgent attention and resolution. Therefore, it is essential for the Republic of Azerbaijan to develop both long-term and short-term climate projections, gather information regarding sea level fluctuations, and implement a series of strategic measures.

Keywords: Caspian Sea, Climate Change, Water Level, Prognosis, Marine Ecosystem, Ecological Condition, Pollution, Wastewater

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1. Introduction

Many researchers suggest studying the Caspian Sea as a large aquatic ecosystem that encompasses the Volga River basin and other nearby river systems. Results from numerous scientific studies conducted so far indicate that the water level of the Caspian Sea is currently continuing to decrease, with projections suggesting that this decrease could reach 5–7 meters in the coming decades. This process is expected to have a particularly significant impact on the southern part of the Caspian, potentially adversely affecting the ecology and habitat of these areas, especially in Azerbaijan [1,2,5,6].

One of the distinctive characteristics that sets the Caspian Sea apart from other large water bodies is its water level, which fluctuates continuously across a wide range. According to historical, archaeological, paleogeographical, and other data, the amplitude of the Caspian Sea's level fluctuations over the past 3,000 years has been close to 15 meters. As stated in the literature, in 1837, academician C. Lents initiated observations on the changes in the Caspian Sea's level on the Baku coastline. Between 1837 and 1982, the highest recorded water level was -25.2 meters in 1882, while the lowest level, recorded in 1977, was -29.0 meters, resulting in an overall decrease of 3.8 meters. However, despite the general decline in the water level, increases have also been observed, as the water level rose by 2.5 meters between 1978 and 1995, reaching -26.5 meters. After this period, the Caspian Sea's level remained relatively stable, with only a slight decrease observed between 2016 and 2020 (0.2 meters). It is projected that the Caspian Sea's level will continue to decrease over the next 25 years, approximately until 2050. According to observations from June 2024, the average water level of the Caspian Sea was recorded at -28.67 meters in the Baltic system, with the Central Caspian at -29.19 meters and Kazakhstan at -28.85 meters [8].

Researchers explain the cause of these level changes through two factors, classifying them into two groups: those who prioritize the climatic factor and those who emphasize the geological factor. Most geologists, however, tend to

prioritize the climatic factor. Nevertheless, both geological and anthropogenic factors should also be taken into account in these processes. The climatic factor influences the Caspian's water balance, particularly in the inflow (waters from rivers flowing into the sea and precipitation) and outflow (evaporation from the surface) components. Disruptions to the water balance lead to changes in the sea level. Scientists note that since the fluctuations in the sea's level are primarily dependent on climatic processes, and due to the current inability to accurately predict future climate forecasts, it is impossible to make a precise long-term projection of these fluctuations, and the existing forecasts are probabilistic in nature [6,7].

Considering these factors, it is necessary to investigate the impacts of various factors, including the increase in evaporation from the sea's surface due to rising temperatures, the condition of rivers flowing into the Caspian, the level of precipitation, and the effects of solar radiation on the water level in the Caspian Sea basin.

2. Experimental detail

The Caspian Sea, rich in natural resources, is the largest enclosed water body on Earth. Due to its size, hydrometeorological regime, and distinctive fauna, it closely resembles a sea rather than a lake. Prof. Q. Gül (1956) discovered up to 30 different names for the Caspian. These names are primarily linked to cities that once existed along the sea's shores, as well as the names of peoples who lived in the coastal regions. Over time, various peoples have assigned nearly 70 different names to the Caspian Sea.

The formation of the Caspian Sea began approximately 10 million years ago, during which time it underwent several phases of transgression and regression, with its water level fluctuating between different heights. The amplitude of these fluctuations in the Caspian Sea's water level is close to 300 meters. During certain periods, the Northern Caspian was completely dry, and part of the Central Caspian also dried up, with only the Southern Caspian, specifically the Lankaran Depression, remaining filled with water.

It is important to note that the physical parameters of the sea change depending on its water level. The Caspian's coastline spans approximately 4,800 km (6,000 km including islands), and five countries are located along this coastline. The length of the coastline held by these countries is as follows: the Republic of Azerbaijan – 813 km, Kazakhstan – 1,422 km, the Islamic Republic of Iran – 728 km, the Russian Federation – 747 km, and Turkmenistan – 1,035 km (Figure 1).

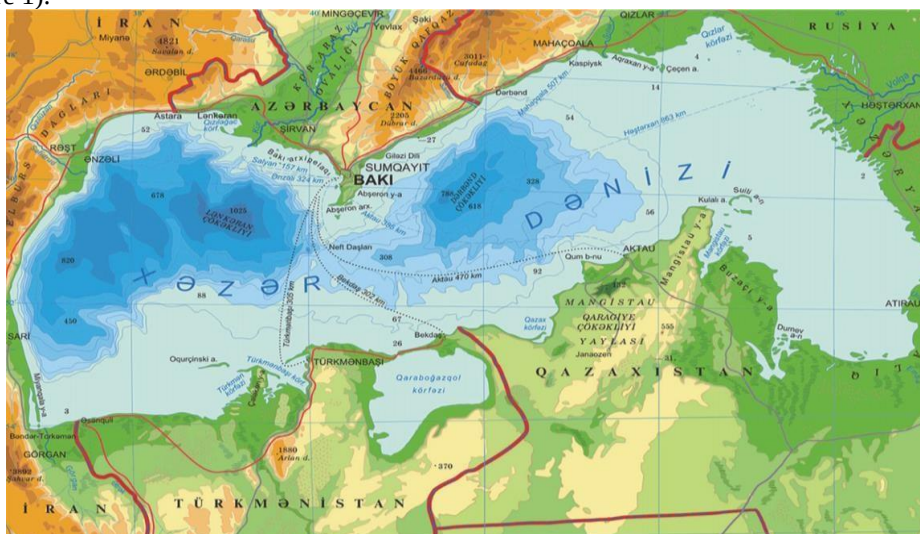


Figure 1. Map of the Caspian Sea.

The Caspian Sea's drainage basin covers an area of 3.1 to 3.5 million km², accounting for 10% of the world's enclosed water bodies. This area stretches 2,500 km from north to south and approximately 1,000 km from west to east. The countries within this region include Azerbaijan, Georgia, Turkey, Kazakhstan, Iran, Russia, Turkmenistan, and Armenia.

Researchers believe that the primary cause of the Caspian Sea's level fluctuations is the climatic factor. Given its vast drainage basin and geographical location, the Caspian Sea can be considered an important climatic and ecological indicator for large areas of the Earth.

Table 1. Modern Physical Parameters of the Caspian Sea

Area	General	371000-429140 km ²
	Northern part	92750-126596 km ²
	Central part	133560-151626 km ²
	Southern part	144690-151018 km ²
Water collection basin	General	3,77 mln.km ²
	USSR (former)	3,5 mln.km ²
	Iran	264 min km ²
Volume	General	75600 km ³
	Northern part	900 km ³
	Central part	26400 km ³
	Southern part	48300 km ³
Length	North to south	1204-1225 km
	West to east	330 km (orta)
Average/max/depth	Northern part	3,5m/20m
	Central part	190m/788
	Southern part	300m/1025m
Length of the coastline	Total part	4800 km,
	With islands	6000 km-ə qədər
	Azerbaijan	813km
	Kazakhstan	1422 km
	Iran	728 km
	Russia	747 km
	Turkmenistan	1035 km
Salinty	Total (central)	12,85 ‰
	Northern part	5-10 ‰
	Central part	12,7 ‰
	Southern part	13,0 ‰

More than 130 rivers flow into the Caspian Sea, with the Volga being the largest river discharging into it. The drainage basin of this river, within the limits of surface currents, covers an area of 1.36 million km².

The largest rivers in the Azerbaijani part of the Caspian Sea are as follows: the Kura (annual flow: 16 km³), Lankaran River, Sumgayit River, Velvelechay, Qudyalchay, and Astara River (on the Azerbaijan-Iran border). The largest rivers along the Azerbaijan-Russia border are the Samur (2.7 km³) and the Terek (8.5-11.4 km³) in Russia.

Table 2. Characteristics of the Major Rivers Flowing into the Caspian Sea

River name	Delta area, km ²	Water flow, km ³		Flow of imports, million tons	
		At the beginning of the delta	At the Sea	At the beginning of the delta	At the Sea
Ural	500	7	6,6	2,7	2,7
Volga	18000	243	233	14	6
Terek	8900	8,9	8,4	15,1	11,9
Sulak	70	4,4	4	13,2	1,6
Samur	80	1,63	1,63	4,7	
Kura	204	7,8	15,5	39,7	17,1
Safidrud	1800	4,67	3,93	31	31
Kharaz	315	0,96	0,67	2,4	1,7

Gorganrud	630	0,49	0,39	3,1	2,5
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The Caspian Sea is located in a seismically active region. In 1895, an earthquake with a magnitude of 8.2 on the Richter scale (11-12 on the 12-point scale) occurred in Krasnovodsk. Experts believe that intensive oil exploration and extraction in the Caspian accelerates the movement of tectonic plates, thereby activating seismic activity.

The Caspian coastline is one of the key regions in the Eurasian area where numerous aquatic and shorebird species thrive. The Caspian Sea is home to three reserves: the Astrakhan Reserve, the Kyzyl-Agach Reserve, and the Caspian Reserve. The Caspian Sea holds a wealth of natural resources, including hydrocarbons, biological resources, recreational and health tourism potential, and mineral salts [6].

The Caspian Sea, the largest lake in the world, stands out due to its geographical location, diverse natural resources, and unique characteristics, constantly attracting the attention of both foreign and local investors. Over many years, industrial and household wastewater, as well as the waste from the oil industry, maritime transport, and tourism facilities along the coastline, have negatively impacted the Caspian's ecosystem.

Climate change has significant and diverse effects on the status of water resources, as well as the coastal and marine ecosystems of the Caspian Sea basin. In recent decades, there is increasing evidence that climate and environmental changes have already impacted the global hydrological cycle, for instance, through changes in seasonal river flows in the Caspian Sea basin. Notably, between 2005 and 2020, the water level of the Caspian Sea has decreased by 133 cm. [8].

American water resources expert Sifan Koriçe emphasized the importance of regional and global monitoring of large lakes, noting that their impact on global climate has not been adequately assessed.

The effect of climate change on the dynamics of the Caspian Sea's water level, the socio-economic and ecological problems caused by these level changes in the coastal zone of our country, the increasing natural disasters resulting from climate change, and other related issues remain highly relevant [9-22].

According to data from the Hydrometeorological Services of the Caspian Sea riparian countries and the Federal Hydrometeorological and Environmental Monitoring Service of Russia, the average water level of the Caspian Sea in 2020 decreased by approximately 3 cm, reaching -28.23 meters below sea level (BS), compared to the average level of -28.20 meters BS in 2019. Due to the reduced flow of the Volga River, it is projected that the average water level of the Caspian Sea in 2021 will be 12–17 cm lower than in 2020. Anthropogenic activities, including economic activities, contribute to the intensification of climate change effects, which, in turn, pose an additional threat to hydrological flow regimes that play a crucial role in the resilience of aquatic ecosystems. Projections suggest that by the end of the century, from 2071 to 2100, the average annual temperature of the Earth's surface in Europe will rise between 1.0°C and 4.5°C compared to the 1971-2000 period. Furthermore, calculations indicate that under a global warming scenario of 1.5-2.0°C, especially in Southern Europe, the availability of water will continue to decrease compared to pre-industrial levels. [11].

In recent years, Azerbaijan has faced acute water scarcity, and the index of annual water resource usage (AWRI) has exceeded 40%. According to calculations, between 2017 and 2019, this index was 72.3%. Due to the new climate conditions, only 25% of the total precipitation contributes to the formation of internal river flow in the country. Between 2000 and 2020, the average annual river flow of the Kura River decreased by more than 50% (Figure 2). In recent years, negative trends have been observed in the formation of both abiotic and biotic factors of natural complexes within the Kura River basin.

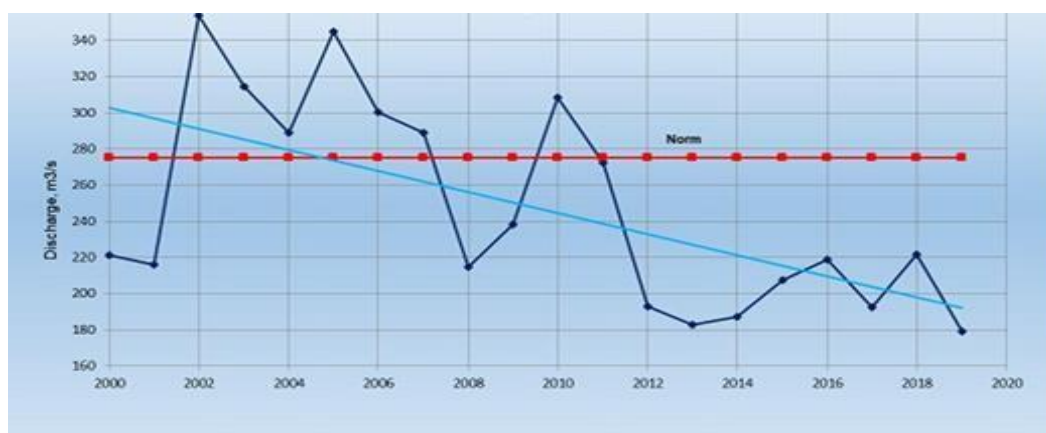


Figure 2. Average Annual River Flow of the Kura River from 2000 to 2020.

It is particularly noteworthy that, for the first time in Azerbaijan, due to a sharp decrease in the river's water level during 2020-2021, sea water from the Caspian Sea entered the Kura River up to 60 km near the city of Salyan. In the summer of 2021, comprehensive marine ecological studies were conducted on the Azerbaijani shelf of the Caspian Sea. Due to unusually high air temperatures, the most significant increase in surface water temperature over the past decade was observed. In the Central Caspian, the temperature ranged from 28-29°C, and in the Southern Caspian, it

ranged from 30-31°C. Compared to 2011, the surface water temperature had increased by 4.0–4.5°C (Figure 3). Additionally, in the context of this abnormal rise in water temperature, a decrease in dissolved oxygen was recorded, with the concentration dropping from 7-8 mg/l to 6 mg/l, compared to observations from previous decades. In the rising zone of the Southern Caspian, as the surface water temperature increased, the anomalous growth of cyanobacteria colonies was observed at depths of 50–100 meters along the Astara-Kura-Kosa route. These colonies have harmful effects on plankton and lead to a reduction in the amount of dissolved oxygen in the sea water.[8].

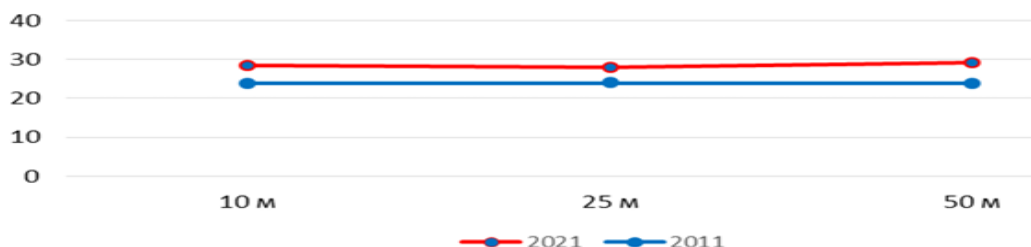


Figure 3. Surface Water Temperature in the Central and Southern Caspian in the Summer of 2011 and 2021.

In recent years, climate change and the anomalous warming occurring in the Caspian region have led to a significant decrease in river flow and an increase in sea water temperature, resulting in considerable degradation of the marine ecosystem. This, in turn, further increases the risk of losing endemic species and the decline of biodiversity in the Caspian Sea in the near future [8].

According to researchers' projections, between 2010 and 2100, the areas marked in red in the Caspian Sea basin will dry up (Figure 4).

Along with the fluctuations in the Caspian Sea's water level, several ecological problems exist in the coastal zones of our country [1,3,4]:

Pollution caused by waste from cities and industrial facilities located along the coast, which flows into the Caspian Sea via rivers;

Pollution related to oil extraction and transportation in the sea;

Pollution from submerged sources in the coastal zone due to the rising water level of the Caspian Sea.

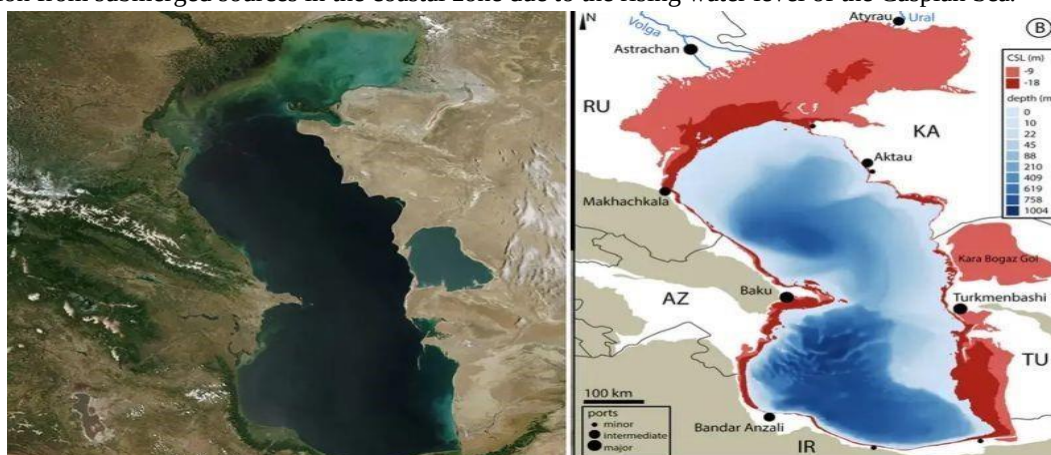


Figure 4. Areas that will Dry Up in the Caspian Sea Basin (2010-2100).

-Oil and petroleum product spills during the transportation and transfer of floating vessels (tankers);

-Pollution from fertilizers used in agriculture, washed into the rivers and canals through rainfall and irrigation water, flowing into the sea;

-Pollution caused by various factors in shipping, mainly ballast water, domestic sewage, and sub-floor waters (oil-bearing layers), as well as waste generated during ship repairs in shipyards.

Among these pollution sources, the primary one is the rivers flowing into the Caspian, with the majority of the pollution coming from the Volga River. Numerous small and large rivers flow into the Caspian Sea from the territory of the Republic of Azerbaijan, such as the Qudyalchay, Sumgayitchay, Vilashchay, and Lankaran rivers, which are partially contaminated with domestic and fecal wastewaters. The Kura River, which enters Azerbaijan from Georgia, is polluted in Georgia, and, similarly, the Araz River, which merges with the Kura River near the Sabirabad region, is further polluted after passing through the territory of Armenia. Another contributing factor to Caspian Sea pollution is the improper management of waste generated by enterprises and facilities located near natural flows and along the coastline. As a result, the Caspian Sea is exposed to anthropogenic impacts, leading to the destruction of aquatic organisms. A major cause of pollution in the Caspian is also the presence of oil stains along the coastal zone and

beaches, the ineffective operation of treatment facilities in large industrial plants, and the discharge of sub-floor waters (from oil-laden layers) from floating vessels (ships) into the sea.

-To prevent oil pollution in the Caspian Sea, the following measures are essential:

-Preventing the discharge of untreated wastewater from large industrial enterprises (oil and gas extraction plants, oil refineries, chemical industry plants, etc.) into Caspian-affiliated streams;

-Replacing outdated equipment in oil and petrochemical enterprises with modern technology that meets current requirements;

-Replacing long-serving hydro-technical facilities operating in the open sea and under the control of oil and gas extraction facilities with those that meet modern standards;

-Taking immediate action to address oil pollution when detected;

-Achieving progress in environmental protection in the Caspian Sea.

After gaining independence, Azerbaijan adopted several significant national regulations aimed at ensuring the ecological security of the Caspian Sea. In 2001, the Ministry of Ecology and Natural Resources of Azerbaijan established the Caspian Complex Ecological Monitoring Department to ensure the ecological safety of the Caspian Sea sector belonging to Azerbaijan. The Department operates under the Constitution of the Republic of Azerbaijan, the Ministry's regulations, orders, decrees, and ratified international treaties, agreements, and conventions, organizing and conducting monitoring in the Azerbaijan sector of the Caspian Sea, along its coast from north to south, including beach areas and facilities operating on the sea, floating vessels, etc. The objective of monitoring is to collect data and analyze the impact of all types of waste (liquid, gas, solid, etc.) generated by enterprises that impact the Caspian Sea through direct or indirect channels (collectors, channels, rivers, etc.), assessing the effect on the sea water, ground, flora, fauna, and other aquatic objects, and to implement measures to minimize these impacts [5,8]. By the decree of the President of the Republic of Azerbaijan on September 28, 2006, the "Comprehensive Action Plan for Improving the Ecological Situation in the Republic of Azerbaijan for 2006-2010" was approved. This plan aimed at restoring the environmental condition of areas such as the Baku Bay, Bibiheybat zone, the surroundings of Baku International Airport, the lakes on the Absheron Peninsula, oil-polluted lands, areas submerged by oil layers, and other areas polluted by industrial waste. Based on the President's directive, financial resources were allocated for the improvement of waste management in Baku and Sumgayit cities, as well as the implementation of other measures to address these environmental issues in a timely manner.

In the direction of protecting the Caspian Sea's environment, 17 stations were established, equipped with modular-type wastewater treatment facilities capable of processing 7,900 cubic meters of wastewater daily. All anthropogenic impacts on the Caspian Sea through various water channels have been inventoried, mapped, and serious measures have been taken to address these issues [5].

To ensure the timely detection and elimination of ecological problems in the Caspian Sea, the Caspian Complex Ecological Monitoring Department of the Ministry of Ecology and Natural Resources organizes and carries out monitoring activities in the Azerbaijani sector of the Caspian Sea, along its 955 km coastline from north to south, including beach areas, as well as facilities operating in the sea, floating vessels, and other relevant sites. During monitoring, all types of waste (liquid, gas, solid) generated by enterprises that affect the Caspian Sea through direct or indirect channels (collectors, canals, rivers, etc.) are analyzed. The impact of these waste materials on the sea water, ground, flora, fauna, and other aquatic environments is thoroughly examined, data is collected, and measures are implemented to minimize these impacts [5,8].

Samples collected from monitoring points are submitted to the laboratories of the Caspian Complex Ecological Monitoring Department. In the laboratory, analytical, biological, ecotoxicological, and microbiological analyses are conducted on the samples of water, bottom sediment, sludge, and drilling mud obtained during the monitoring [5,8]:

- Analytical Research – Physical-chemical and chemical analyses are conducted on the samples.
- Ecotoxicology – Ecotoxicological analyses are performed on water samples, drilling mud, sludge, and bottom sediments using the test organisms *Daphnia magna* and *Pontogammarus maeoticus*.
- Hydrobiology – Biological analyses are conducted on water samples to determine the quantity of Total Coliform Bacteria and the conditionally pathogenic bacterium *Escherichia coli* (*E. coli*), as well as to study the number and biomass of zooplankton and zoobenthos organisms. Additionally, biological analyses are performed on active sludge and water samples to assess the efficiency of treatment facilities. [8].

3. Conclusion

Among the current issues facing the Caspian Sea, the ecological condition, changes in water level, and its overall status are of significant importance. In relation to climate variability, the Caspian Sea is confronted with difficult and urgent problems that need to be addressed. Therefore, long- and short-term climate projection scenarios must be developed for the Republic of Azerbaijan, information on the frequency of stormy days, expected annual temperature

indicators, and sea level fluctuations must be gathered and generalized, and the following measures must be implemented:

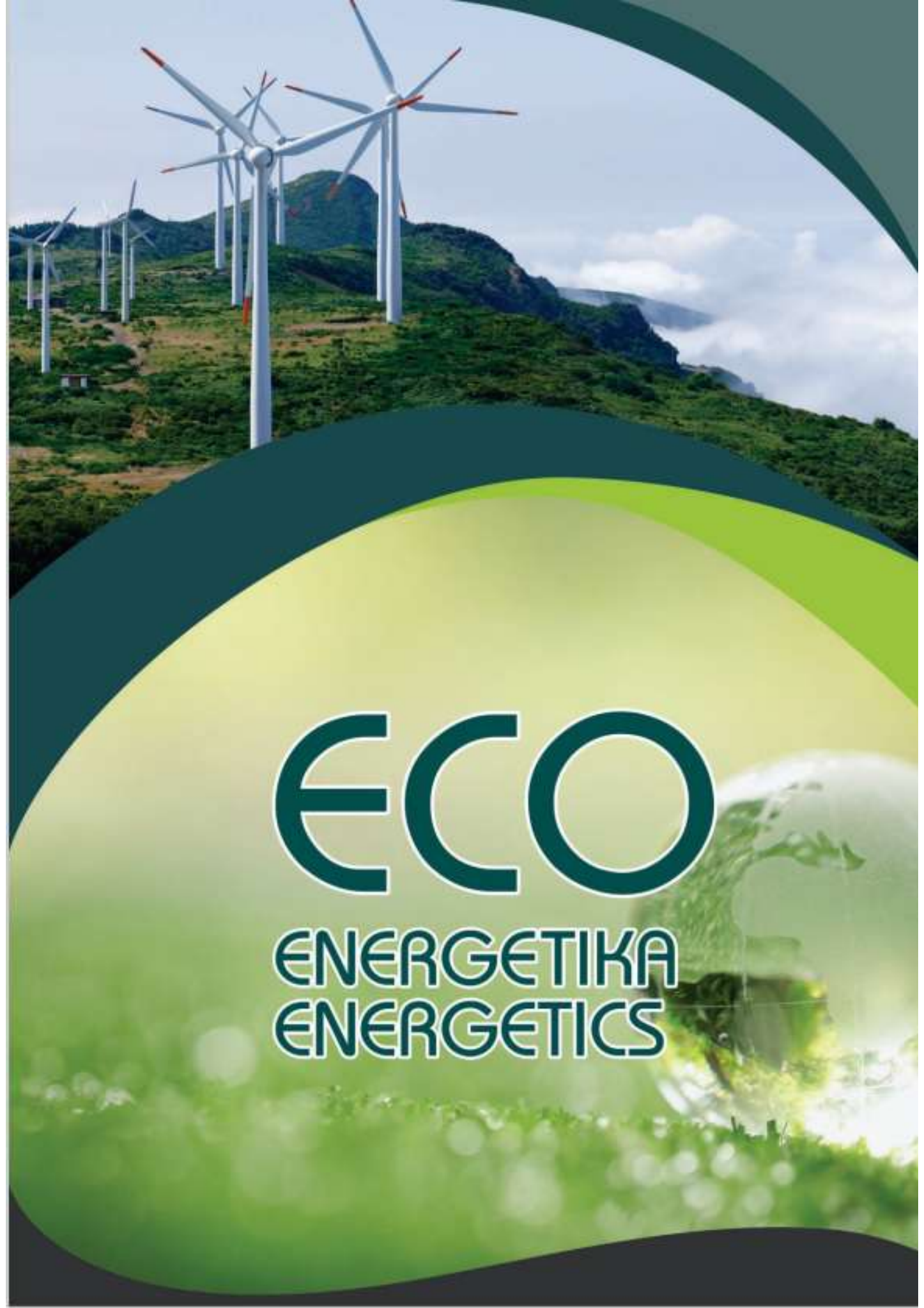
- The application of legal and economic standards to protect the natural environment from anthropogenic, technogenic, and other events;
- The implementation of technical and economic activities to eliminate potential negative events;
- The application of international cooperation to eliminate anthropogenic and other impacts;
- The establishment of a unified monitoring system for the organization of measures to improve the ecological condition of the Caspian;
- Measures to completely prevent the discharge of industrial waste into the sea by Caspian coastal states;
- Improvement of the operation of offshore oil and gas extraction facilities;
- The use of equipment that meets international standards for wastewater treatment;
- The use of oil-oxidizing bacteria in water bodies and areas polluted with oil products on land;
- The implementation of activities to develop ecological culture, education, and ethics among the population.

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The background of the entire image is a photograph of a wind farm. Several white wind turbines with three blades each are positioned on a green, grassy hill. The sky is blue with some white clouds. The image is framed by a dark green curved border at the top and bottom. A large, light green semi-circular shape is overlaid on the bottom half of the image, containing the text.

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